P1 Practice: Armijo, weak Wolfe and strong Wolfe intervals

Data

$f(x_1, x_2) = 10x_1^2 + x_2^2$, $x_0 = (1, 1)^T$, $\nabla f(x) = (20x_1,\, 2x_2)^T$, so

$$p_0 = -\nabla f(x_0) = (-20, -2)^T, \qquad x_0 + \alpha p_0 = (1 - 20\alpha,\; 1 - 2\alpha)^T,$$

with $c_1 = 10^{-4}$ and $c_2 \in \{0.9,\; 0.1\}$.

The function ϕ

$$\phi(\alpha) = 10(1 - 20\alpha)^2 + (1 - 2\alpha)^2 = 11 - 404\,\alpha + 4004\,\alpha^2, \qquad \phi'(\alpha) = -404 + 8008\,\alpha.$$

Hence $\phi(0) = 11$, $\phi'(0) = -404 < 0$ (descent direction), and ϕ is a convex parabola with minimizer

$$\alpha^* = \frac{404}{8008} = \frac{101}{2002} \approx 0.050450.$$

Since ϕ is quadratic, write $\phi(\alpha) = \phi(0) + \phi'(0)\alpha + \frac{1}{2}a\alpha^2$ with $a = 8008$ and $\alpha^* = -\phi'(0)/a$.

Armijo

$$\phi(\alpha) \le \phi(0) + c_1\alpha\phi'(0) \iff 4004\,\alpha^2 \le 404(1 - c_1)\,\alpha.$$

For $\alpha > 0$ divide by α:

$$\alpha \le \frac{404(1 - c_1)}{4004} = 2(1 - c_1)\,\alpha^*.$$

With $c_1 = 10^{-4}$:

$$\boxed{0 < \alpha \le \frac{91809}{910000} \approx 0.100889}$$

Weak curvature

$$\phi'(\alpha) \ge c_2\phi'(0) \iff -404 + 8008\alpha \ge -404c_2 \iff \alpha \ge \frac{404(1 - c_2)}{8008} = (1 - c_2)\,\alpha^*.$$

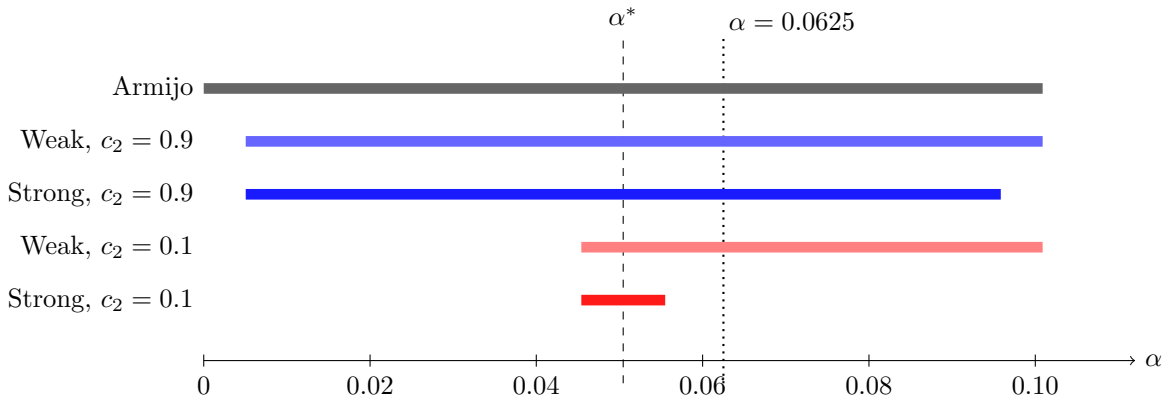
Strong curvature

$$|\phi'(\alpha)| \le c_2|\phi'(0)| \iff -404c_2 \le -404 + 8008\alpha \le 404c_2 \iff (1 - c_2)\,\alpha^* \le \alpha \le (1 + c_2)\,\alpha^*.$$

Intervals

Weak Wolfe = Armijo ∩ weak curvature; strong Wolfe = Armijo ∩ strong curvature. Because $c_2 < 1 - 2c_1 = 0.9998$ in both cases, $(1 + c_2)\alpha^* < 2(1 - c_1)\alpha^*$, so the Armijo bound does not cut the strong interval.

	$c_2 = 0.9$	$c_2 = 0.1$
Armijo	$(0,\ 0.100889]$	$(0,\ 0.100889]$
Weak Wolfe	$\left[\frac{101}{20020},\ \frac{91809}{910000}\right] \approx [0.005045,\ 0.100889]$	$\left[\frac{909}{20020},\ \frac{91809}{910000}\right] \approx [0.045405,\ 0.100889]$
Strong Wolfe	$\left[\frac{101}{20020},\ \frac{1919}{20020}\right] \approx [0.005045,\ 0.095854]$	$\left[\frac{909}{20020},\ \frac{101}{1820}\right] \approx [0.045405,\ 0.055495]$



Check against the backtracking step

Backtracking (V1) accepted $\alpha = 0.0625$, with $\phi'(0.0625) = 96.5$.

- $0.0625 \leq 0.100889$: Armijo holds.
- $0.0625 \geq 0.005045$ and $0.0625 \geq 0.045405$: weak Wolfe holds for both c_2.
- $0.0625 \leq 0.095854$: strong Wolfe holds for $c_2 = 0.9$.
- $0.0625 > 0.055495$: strong Wolfe fails for $c_2 = 0.1$.

This agrees with V1. Note that the strong Wolfe interval is always centred at α^* with half-width $c_2\alpha^*$, so a small c_2 forces the step close to the exact minimizer.